IMEN266 Operations Research II
Chapter 7. Renewal Theory

Fall, 2026



INDUSTRIAL AND MANAGEMENT
ENGINEERING, POSTECH

Outline

1. Renewal Process
2. Limiting behavior
3. Wald's equation
4. Renewal reward process
5. Regenerative process

How to study this chapter (2026 format)

- **Slides** (this file): concepts, class-problem statements, and the worked derivations developed in class. No transcription needed — definitions, theorems and proofs also live in the Companion.
- **Notebooks** (CP9 renewal process, renewal function, limit theorems, Wald · CP10 renewal reward, age/residual life, inspection paradox, regenerative and alternating renewal processes): in class the `INCLASS` version (fill the `TODO`s, `check()` answers instantly); at home the full version with solutions and experiments.
- **Companion** (gap-fill $\rightarrow$ full): all definitions and theorems + 8 core proofs with blanks, checkpoints, AI study cards.
- **Homework 4**: A exam-style (no AI first) · B critique an AI solution · C stress-test the rate of convergence and the insensitivity claims by simulation · D prompt log.
- Class rhythm (75 min): quiz $5'$ $\rightarrow$ concepts $\rightarrow$ live notebook $\rightarrow$ wrap-up. Ch. 7 = weeks 9–11.
- Ground truth = analytic $\leftrightarrow$ simulation agreement, not AI confidence. Materials: PLMS and GitHub `imen266-2026`.

Renewal Process

- Recall the Poisson process
- What if inter-event distribution is not exponential?
- The renewal process is a generalization of the Poisson process by relaxing the exponential assumption of inter-event times.

Definition

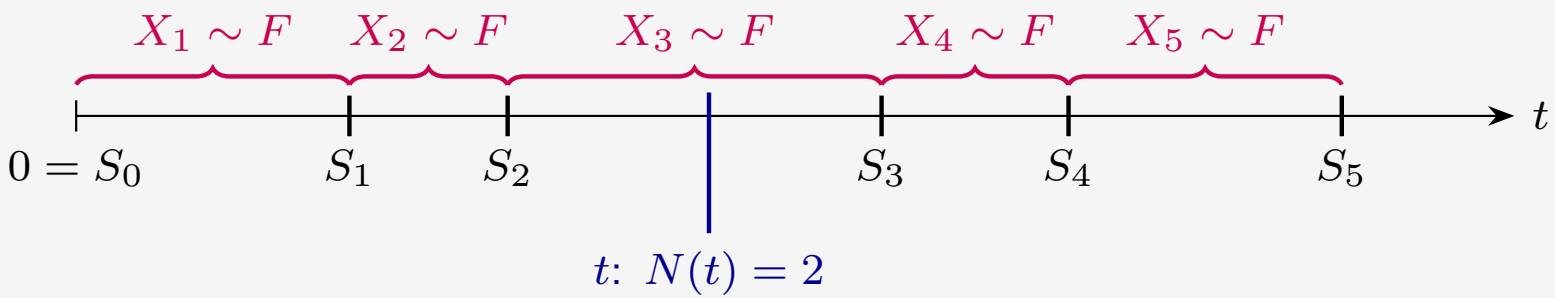
Suppose we have a sequence of iid non-negative random variables, $X_1, X_2, \ldots$. Then, the counting process $\{N(t), t \geq 0\}$ is called a renewal process.

▷ Companion §1.1: S_n, $N(t)$, and the key identity $N(t) \geq n \Leftrightarrow S_n \leq t$ (Proof 1).

▷ Recall: the Poisson process is the counting process whose inter-event times are iid *exponential*; a renewal process keeps "iid" and drops "exponential".

Renewal Process

The objects: gaps X_n, renewal epochs S_n, the count $N(t)$



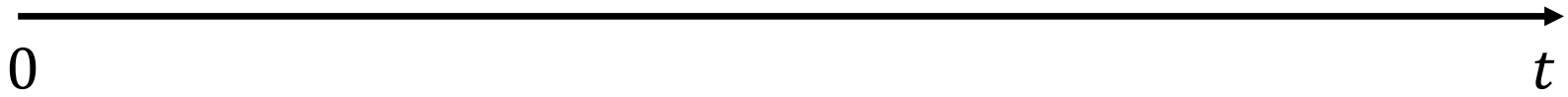
$X_1, X_2, \ldots$ iid with distribution F, $\mu = E[X_1]$; $S_n = X_1 + \cdots + X_n$ is the time of the n-th renewal ($S_0 = 0$); $N(t) = \max\{n : S_n \le t\}$ counts the renewals in $(0, t]$. The key identity, read off the picture: $N(t) \ge n \iff S_n \le t$.

Two standing assumptions

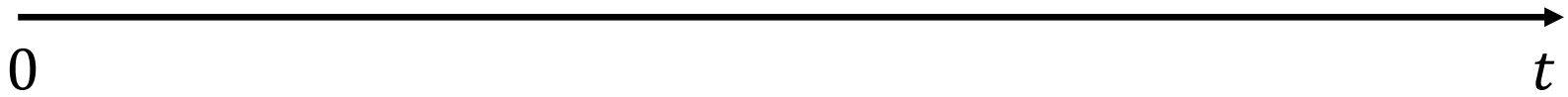
$F(0) = P(X_n = 0) < 1$, so $\mu > 0$ and, by the strong law, $S_n/n \to \mu > 0$: the epochs $S_n \to \infty$ and $N(t) < \infty$ for every t (only finitely many renewals in a finite interval). The Poisson process is the special case $F = \exp(\lambda)$.

Why renewal?

- Poisson process

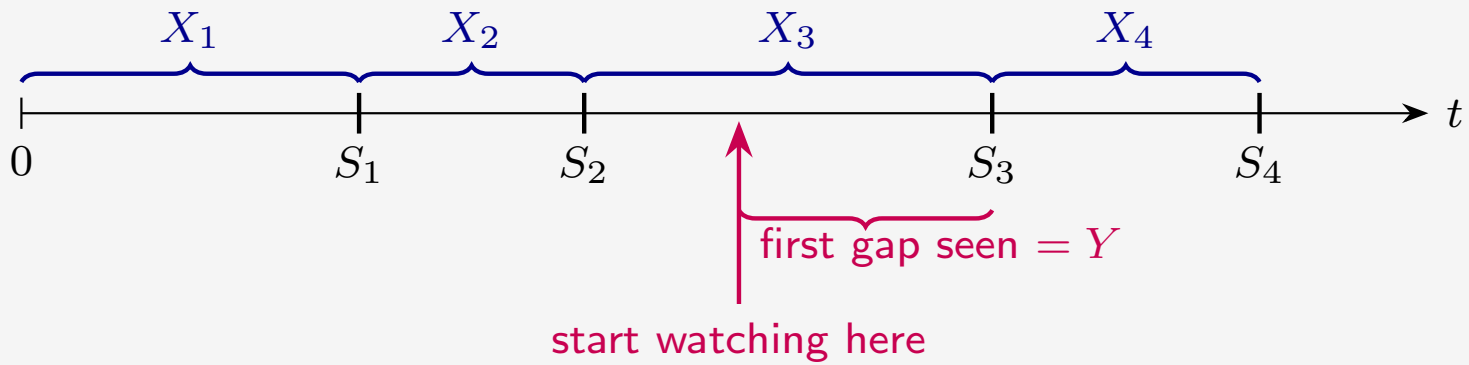


- Renewal process



Why renewal?

Where does the process restart?



Poisson process ($X_n \sim \exp(\lambda)$): start watching at *any* time — by memorylessness the first gap seen is again $\exp(\lambda)$, independent of the past: the Poisson process is renewed at every time point you choose.

Renewal process ($X_n \sim F$): from a non-event time the first gap seen is the *remaining* part of an interval; it does not have distribution F (it tends to be longer — inspection paradox). Only at an event time does the future look like a fresh copy: a renewal process is renewed at its event times $S_1, S_2, \ldots$ only.

▷ Hence the age $A(t)$ and the remaining time $Y(t)$ at a non-renewal time t.

Distribution of $N(t)$

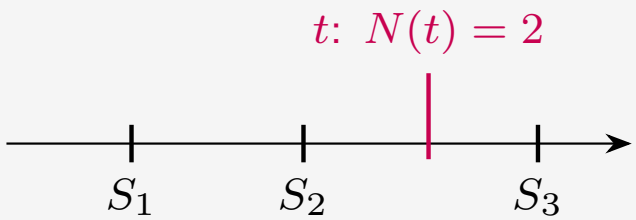
$$
\begin{aligned}
P\left(N(t)=n\right) &= P\left(N(t)\geq n\right) - P\left(N(t)\geq n+1\right) \\
&= P\left(S_n \leq t\right) - P\left(S_{n+1}\leq t\right) \\
&= P\left(\sum_{i=1}^{n} X_i \leq t\right) - P\left(\sum_{i=1}^{n+1} X_i \leq t\right) \\
&= F_n(t) - F_{n+1}(t).
\end{aligned}
$$

- Example: Poisson process

▷ Companion Proof 1: $P(N(t)=n) = F_n(t) - F_{n+1}(t)$; for exponential gaps this recovers $e^{-\lambda t}(\lambda t)^n/n!$.

Distribution of $N(t)$

Why $P(N(t) \geq n) = P(S_n \leq t)$, and what F_n is



"at least n renewals by time t" $\Leftrightarrow$ "the n-th renewal happened by t":
$\{N(t) \geq n\} = \{S_n \leq t\}$.
$F_n(t) = P(S_n \leq t)$ is the
n-fold convolution of F: $F_1 = F$,
$F_{n+1}(t) = \int_0^t F_n(t-x)\, dF(x)$.

Example: exponential gaps give back the Poisson pmf

$S_n \sim \text{Erlang}(n, \lambda)$, so $F_n(t) = 1 - \sum_{i=0}^{n-1} e^{-\lambda t} \frac{(\lambda t)^i}{i!}$ ("fewer than n events by t" is the complement), and $P(N(t) = n)$ is

$$F_n(t) - F_{n+1}(t) = \sum_{i=0}^{n} e^{-\lambda t} \frac{(\lambda t)^i}{i!} - \sum_{i=0}^{n-1} e^{-\lambda t} \frac{(\lambda t)^i}{i!} = e^{-\lambda t} \frac{(\lambda t)^n}{n!}.$$

▷ General F: no closed form — CP9 computes F_n numerically and by simulation.

Renewal equation

Lemma

$$m(t) \equiv E\left[N(t)\right] = \sum_{n=1}^{\infty} F_n(t)$$

▷ Companion §1.2 and Proof 2: the renewal function $m(t) = \sum_n F_n(t)$; CP9 computes it three ways.

Renewal equation

Proof of $m(t) = \sum_{n\geq 1} F_n(t)$ — the tail-sum formula

For a nonnegative integer-valued random variable, $E[N] = \sum_{n=1}^{\infty} P(N \geq n)$ (the event $\{N \geq n\}$ is counted once for each $n \leq N$; continuous analogue: $E[X] = \int_0^{\infty} (1 - F(t))\, dt$). Apply it to $N(t)$:

$$m(t) = E[N(t)] = \sum_{n=1}^{\infty} P\big(N(t) \geq n\big) = \sum_{n=1}^{\infty} P(S_n \leq t) = \sum_{n=1}^{\infty} F_n(t)\,.$$

The renewal function

$m(t)$ is the **renewal function** of F: the mean number of renewals in $(0, t]$. It is finite for every t (Companion Proof 1); for the Poisson process $m(t) = \lambda t$, for a general F the sum of convolutions is rarely closed-form — hence the equation on the next slide.

▷ Tail-sum formulas return in Wald's equation and in the age process (§5).

Renewal equation

Lemma (Renewal equation)

$$m(t) = F(t) + \int_0^t m(t-x)dF(x)$$

▷ Companion Proof 2 (final-style): solving the renewal equation — uniform$(0,1)$ by differentiation, hyperexponential by Laplace transform.

Renewal equation

Proof: condition on the first renewal time X_1

$$m(t) = E[N(t)] = E\big[E[N(t) \mid X_1]\big] = \int_0^\infty E[N(t) \mid X_1 = x]\, dF(x).$$

If $X_1 = x \le t$: one renewal at x, then a fresh renewal process on $(x, t]$, so $E[N(t) \mid X_1 = x] = 1 + m(t-x)$. If $x > t$: no renewal by t, so $E[N(t) \mid X_1 = x] = 0$. Hence

$$m(t) = \int_0^t \big(1 + m(t-x)\big)\, dF(x) + \int_t^\infty 0\, dF(x) = F(t) + \int_0^t m(t-x)\, dF(x)\,.$$

What the equation is good for

It determines m from F without any convolution: solve it on a time grid (CP9 §B), by Laplace transforms (Companion Proof 2), or via an ODE (next frame).

▷ After X_1 the process is a fresh copy — the standard first step.

Example: $F \sim$ uniform$(0, 1)$

- Let the inter-event times be uniform$(0, 1)$, so that $F(t) = t$ and $dF(t) = dt$ for $0 \leq t \leq 1$.
- Find the renewal function $m(t)$ for $0 \leq t \leq 1$ from the renewal equation.

▷ Companion Proof 2(c); CP9 §B checks $m(t) = e^t - 1$ against the discretised renewal equation and simulation.

Example: $F \sim$ uniform$(0,1)$

Solve the renewal equation for $0 \le t \le 1$

For $0 \le t \le 1$: $F(t) = t$ and $dF(x) = dx$ on $[0,t]$, so

$$m(t) = t + \int_0^t m(t-x)\,dx = t + \int_0^t m(u)\,du \qquad (u = t - x).$$

Differentiate: $m'(t) = 1 + m(t)$, $m(0) = 0$. Put $g = m + 1$: $g' = g$, $g(0) = 1$, so $g(t) = e^t$ and

$$m(t) = e^t - 1 \qquad (0 \le t \le 1).$$

$m(\tfrac{1}{2}) = e^{1/2} - 1 \approx 0.649$, $m(1) = e - 1 \approx 1.718$ (CP9 §B).

Beyond $t = 1$, and the long-run check

For $t > 1$ the integral only runs over $x \in [0,1]$: $m(t) = 1 + \int_{t-1}^t m(u)\,du$ — piecewise, no longer one exponential (CP9 solves it on a grid). Elementary renewal theorem: $m(t)/t \to 1/\mu = 2$; more precisely $m(t) = 2t - \frac{1}{3} + o(1)$ (Companion §1.2), and $m(1) = 1.718$ is already close to $2 - \frac{1}{3} = 1.667$.

Limit theorem

- Strong law of large numbers

$$\lim_{n\to\infty} \frac{S_n}{n} = \mu = E[X_1] \text{ a.s.}$$

▷ Companion §1.3: almost-sure convergence (one sample path at a time) versus convergence in distribution (CLT).

Limit theorem

What "almost surely" says: one sample path at a time

$S_n/n = \frac{1}{n}\sum_{i=1}^{n} X_i$ is the running average. The strong law states

$$P\left(\left\{\omega : \lim_{n\to\infty} \frac{S_n(\omega)}{n} = \mu\right\}\right) = 1 :$$

for (almost) every realisation ω of the *whole sequence*, the running average converges to μ. Coin flips ($E[X] = \frac{1}{2}$): $\omega_1 = HHHH\cdots$ has running average $\to 1$, $\omega_2 = TTTT\cdots$ has $\to 0$ — such paths exist, but together they have probability 0; a typical path $HHTHT\cdots$ converges to $\frac{1}{2}$.

Contrast: the central limit theorem is about distributions

$$Z_n = \frac{S_n/n - \mu}{\sigma/\sqrt{n}} \;\Rightarrow\; N(0,1)$$ in distribution: the fluctuation of S_n/n around μ has size $\sigma/\sqrt{n}$ and a Gaussian shape. Both are used here: the strong law for long-run *rates*, the CLT for the accuracy of simulation estimates.

Limit theorem

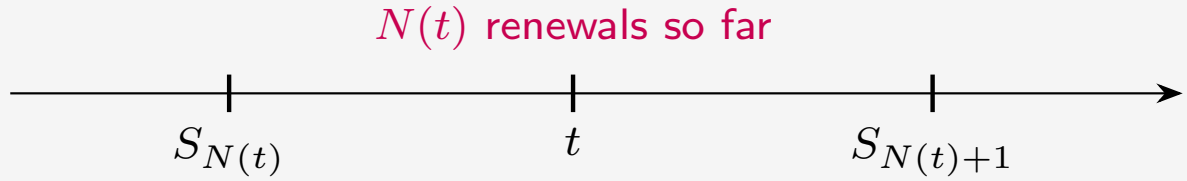
Lemma

$$\lim_{t\to\infty} \frac{N(t)}{t} = \frac{1}{\mu} \text{ a.s.}$$

▷ Companion Proof 3: the sandwich $S_{N(t)}/N(t) \leq t/N(t) < S_{N(t)+1}/N(t)$; the long-run rate of a renewal process is $1/\mu$.

Limit theorem

Proof: sandwich t between two renewal epochs



By definition $S_{N(t)} \le t < S_{N(t)+1}$. Divide by $N(t)$:

$$\frac{S_{N(t)}}{N(t)} \le \frac{t}{N(t)} < \frac{S_{N(t)+1}}{N(t)} = \frac{S_{N(t)+1}}{N(t)+1} \cdot \frac{N(t)+1}{N(t)}.$$

As $t \to \infty$, $N(t) \to \infty$ a.s. (every S_n is finite), so the strong law applies along $n = N(t)$: the left side $\to \mu$ and the right side $\to \mu \cdot 1$. Hence $t/N(t) \to \mu$ a.s., i.e.

$$\lim_{t\to\infty} \frac{N(t)}{t} = \frac{1}{\mu} \quad \text{a.s.} \qquad (\text{rate } 0 \text{ if } \mu = \infty).$$

▷ Only the mean μ enters the rate; the shape of F sets how fast $N(t)/t$ settles (CP9 §C, HW4 Part C).

Elementary renewal theorem

Theorem (Elementary renewal theorem)

$$\lim_{t\to\infty} \frac{m(t)}{t} = \frac{1}{\mu}.$$

▷ Companion Proof 3: $m(t)/t \to 1/\mu$ needs Wald's equation (next section) — a.s. convergence alone does not give convergence of expectations.

Elementary renewal theorem

Why this is not a corollary of the previous lemma

$N(t)/t \to 1/\mu$ almost surely does *not* by itself give $E[N(t)]/t \to 1/\mu$ (a.s. convergence versus convergence of expectations). The proof (Companion Proof 3) uses Wald's equation of the next section:

$$E[S_{N(t)+1}] = \mu\left(m(t)+1\right) \quad \text{and} \quad S_{N(t)+1} > t \quad \Longrightarrow \quad \frac{m(t)}{t} > \frac{1}{\mu} - \frac{1}{t},$$

so $\liminf_t m(t)/t \geq 1/\mu$; the reverse inequality comes from truncating X at a level M.

Sanity check: the Poisson process

$m(t) = E[N(t)] = \lambda t$, so $\dfrac{m(t)}{t} = \lambda = \dfrac{1}{\mu}$ *for every* t — no limit needed.

For a general F the ratio only converges: for uniform$(0,1)$ gaps, $m(1)/1 = e - 1 \approx 1.72$ versus $1/\mu = 2$.

▷ Both: the long-run renewal rate is $1/\mu$, per path and on average.

Class problem 9

1. Kim has a computer mouse that works with a single battery. Whenever the battery is out, Kim replaces it with a new one immediately. Given that the lifetime distribution of the battery is uniform$(50, 100)$, what is the average rate at which Kim replaces the battery in the long run?

▶ Colab notebook: CP9 §1 (battery replacement rate) `[INCLASS] [full]`

Class problem 9

1. Battery replacements: a renewal process with uniform$(50, 100)$ lifetimes

$X_i \sim$ uniform$(50, 100)$ (hours) is the lifetime of the i-th battery; replacement is immediate, so the replacement epochs are $S_n = X_1 + \cdots + X_n$ and $N(t) =$ number of replacements by time t: a renewal process with $\mu = E[X_1] = 75$ hours.

$$\text{replacement rate} = \lim_{t\to\infty} \frac{N(t)}{t} = \lim_{t\to\infty} \frac{m(t)}{t} = \frac{1}{\mu} = \frac{1}{75} \text{ per hour}$$

≈ 0.0133 per hour: about 13.3 batteries per 1000 hours (CP9 §C).

What the answer does and does not depend on

Only the mean lifetime matters: uniform$(50, 100)$, exponential with mean 75 and a deterministic 75-hour battery all give $1/75$. They differ in the *fluctuations*: deterministic lifetimes keep $N(t)/t$ within $1/t$ of $1/75$, the others converge like $\sigma/\sqrt{t}$ (CP9 §C).

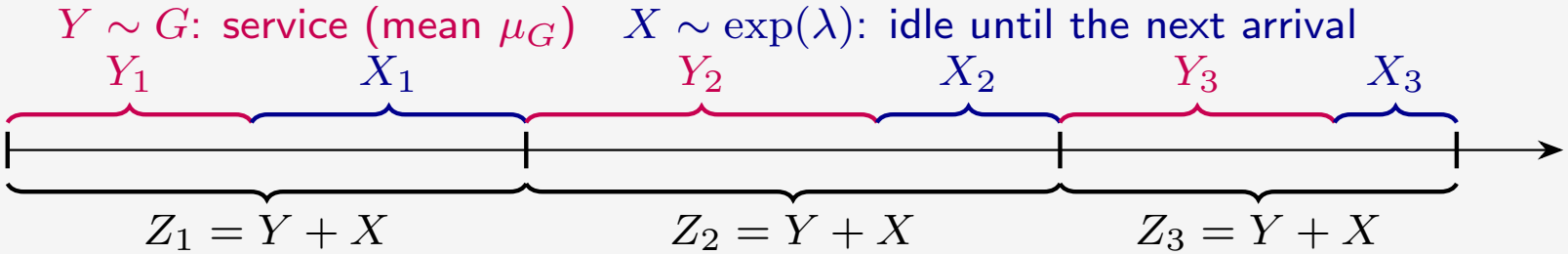
Class problem 9

2. Consider a bank with a single server. Customers arrive according to $PP(\lambda)$. Each customer enters the bank only when the server is free. Otherwise, the customer leaves without entering the bank. The service times of the server follow a general distribution with mean μ_G.
   (a) What is the customers' entering rate to the bank?
   (b) What proportion of arriving customers actually enter the bank?

▶ Colab notebook: CP9 §2 (single-server bank) `[INCLASS] [full]`

Class problem 9

2. Bank with a single server: cycles = service + idle period



A cycle starts when a customer enters (server free): the service Y, then an idle period $X \sim \exp(\lambda)$ until the next arrival (memoryless, whatever happened before), so the cycles $Z_n = Y_n + X_n$ are iid: the entries form a renewal process with $E[Z] = \mu_G + 1/\lambda$.

$$\text{(a) entering rate} = \lim_{t\to\infty} \frac{N(t)}{t} = \frac{1}{E[Z]} = \frac{\lambda}{1+\lambda\mu_G} \text{ a.s.}$$

$$\text{(b) proportion entering} = \frac{\text{entering rate}}{\text{arrival rate } \lambda} = \frac{1}{1+\lambda\mu_G}$$

$\lambda = 4$ per hour, $\mu_G = 10$ min: 2.4 customers per hour enter, i.e. $60\,\%$ of the arrivals (CP9 §C).

Wald's equation

Definition (Stopping time)

The non-negative integer-valued random variable N is said to be a stopping time for a sequence of independent random variables $X_1, X_2, \ldots$ if the event $\{N = n\}$ is independent of $X_{n+1}, X_{n+2}, \ldots$.

- Example: $X_1, X_2, \ldots$ is a sequence of iid random variables with $P(X_i = 1) = P(X_i = -1) = 0.5$.

$$N \equiv \min\{n | X_1 + \cdots + X_n = 1\}.$$

Is N a stopping time?

▷ Companion §1.4: a stopping time may look at the past, never at the future.

Wald's equation

From a fixed number of terms to a random one

Fixed n: $E\big[\sum_{i=1}^{n} X_i\big] = nE[X_1]$. Random N: $E\big[\sum_{i=1}^{N} X_i\big] = ?$ If N is *independent* of the sequence: $E[N]E[X_1]$ (condition on N). Wald's equation keeps this formula when N may depend on the X_i's, provided "stop at n" looks only at $X_1, \ldots, X_n$:

$\{N = n\}$ is determined by $X_1, \ldots, X_n$ (a stopping time).

The example: $N = \min\{n : X_1 + \cdots + X_n = 1\}$ with $P(X_i = \pm 1) = \frac{1}{2}$

$\{N = n\}$ = "the partial sums stayed below 1 up to step $n-1$ and hit 1 at step n" — determined by $X_1, \ldots, X_n$: **yes, a stopping time**. (Not one: "the last $n \le 10$ with $X_n = 1$".)

▷ Caution: $E[N] = \infty$ here, so Wald does *not* apply: $E[S_N] = 1 \neq E[N]E[X_1]$.

Wald's equation

Theorem (Wald's equation)

If $X_1, X_2, \ldots$ is a sequence of iid random variables with finite expectation and if N is a stopping time for this sequence such that $E[N] < \infty$, then

$$E\left[\sum_{n=1}^{N} X_n\right] = E[N]E[X].$$

▷ Companion Proof 4: Wald's equation via the indicators $I_n = \mathbf{1}\{N \geq n\}$, which depend only on $X_1, \ldots, X_{n-1}$.

Wald's equation

Proof (1): write the random sum with indicators

$$\sum_{n=1}^{N} X_n = \sum_{n=1}^{\infty} X_n \, I_n, \qquad I_n = \begin{cases} 1 & \text{if } n \le N \\ 0 & \text{otherwise} \end{cases} = \mathbf{1}\{N \ge n\}.$$

Claim: I_n and X_n are independent.

$\{N \le n-1\} = \{N = 1\} \cup \{N = 2\} \cup \cdots \cup \{N = n-1\}$, and each $\{N = i\}$ is determined by $X_1, \ldots, X_i$ (stopping time); therefore $\{N \ge n\} = \{N \le n-1\}^c$ is determined by $X_1, \ldots, X_{n-1}$ — note the shift: "$\ge n$" needs only the first $n-1$ terms — and by the independence of the sequence

$$I_n = \mathbf{1}\{N \ge n\} \text{ is independent of } X_n \, .$$

▷ This is the whole trick: a stopping time may use the past, and "still going at step n" is a statement about the past of step n.

Wald's equation

Proof (2): take expectations term by term

$$E\Big[\sum_{n=1}^{N} X_n\Big] = E\Big[\sum_{n=1}^{\infty} X_n I_n\Big] \overset{(i)}{=} \sum_{n=1}^{\infty} E[X_n I_n] \overset{(ii)}{=} \sum_{n=1}^{\infty} E[X_n]\,E[I_n]$$

$$= E[X_1] \sum_{n=1}^{\infty} P(N \geq n) \overset{(iii)}{=} E[X_1]\,E[N]\,.$$

(i) monotone convergence for $X_n \geq 0$ (dominated convergence in general); (ii) independence of X_n and I_n; (iii) tail-sum formula.

Where $E[N] < \infty$ is used

The interchange in (i) for signed X_n requires $\sum_n E|X_n|\,E[I_n] = E|X_1|\,E[N] < \infty$; without it the random sum need not have expectation $E[X]E[N]$ (random-walk example).

▷ Needed: iid terms with a finite mean, and a stopping rule with $E[N] < \infty$.

Wald's equation

Corollary

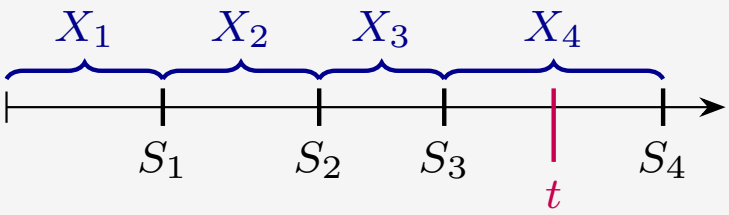
$$E\left[S_{N(t)+1}\right] = E\left[X\right] E\left[N(t)+1\right].$$

▷ Companion Proof 4: $N(t)$ is not a stopping time, $N(t)+1$ is;

$E[S_{N(t)+1}] = \mu\left(m(t)+1\right)$ is the key to Proof 3.

Wald's equation

$N(t)$ is not a stopping time — $N(t)+1$ is



$\{N(t) = 3\} = \{S_3 \le t < S_4\}$ depends on X_4, the term *after* the first three: $N(t)$ looks at the future — **not** a stopping time.

$\{N(t)+1 = 4\} = \{S_3 \le t < S_4\}$ is determined by the first four terms $X_1, \ldots, X_4$:

$N(t)+1$ is a stopping time.

Hence the corollary

$E[N(t)+1] = m(t)+1 < \infty$, so Wald applies to the first $N(t)+1$ gaps:

$$E\left[S_{N(t)+1}\right] = E\left[\sum_{n=1}^{N(t)+1} X_n\right] = E[X_1]\,E[N(t)+1] = \mu\left(m(t)+1\right).$$

▷ By contrast $E[S_{N(t)}] \neq \mu\, m(t)$ in general (CP9 §D checks both).

Renewal reward process

- Each time a renewal occurs, we receive a reward R.
- R_n is a reward earned at n^{th} renewal, independent of each other.

Theorem (Renewal reward process)

Define a stochastic process

$$R(t) = \sum_{n=1}^{N(t)} R_n.$$

If $E[R] < \infty$, $E[X] < \infty$, *then*

$$\text{(a)} \lim_{t\to\infty} \frac{R(t)}{t} = \frac{E[R]}{E[X]} \text{ a.s.}$$

$$\text{(b)} \lim_{t\to\infty} \frac{E[R(t)]}{t} = \frac{E[R]}{E[X]}.$$

Renewal reward process

Reading the theorem

$R(t) = \sum_{n=1}^{N(t)} R_n$ is the **cumulative reward until time** t. The pairs (X_n, R_n) are iid: a cycle's reward may depend on *its own* length, not on the other cycles. Proof of (a):

$$\frac{R(t)}{t} = \frac{\sum_{n=1}^{N(t)} R_n}{N(t)} \cdot \frac{N(t)}{t} \xrightarrow[t\to\infty]{} E[R] \cdot \frac{1}{E[X]} \quad \text{a.s.}$$

by the strong law ($N(t) \to \infty$) and the limit theorem; the partial last cycle adds at most $R_{N(t)+1}/t \to 0$.

$$\text{long-run reward rate} = \frac{E[\text{reward in one cycle}]}{E[\text{length of one cycle}]} \qquad \text{never } E[R/X].$$

The recipe for every class problem (Companion §1.5, Proof 5)

(1) find a **renewal epoch** (the system restarts probabilistically); (2) $E[X] =$ expected cycle length; (3) $E[R] =$ expected reward or cost in one cycle; (4) answer $= E[R]/E[X]$.

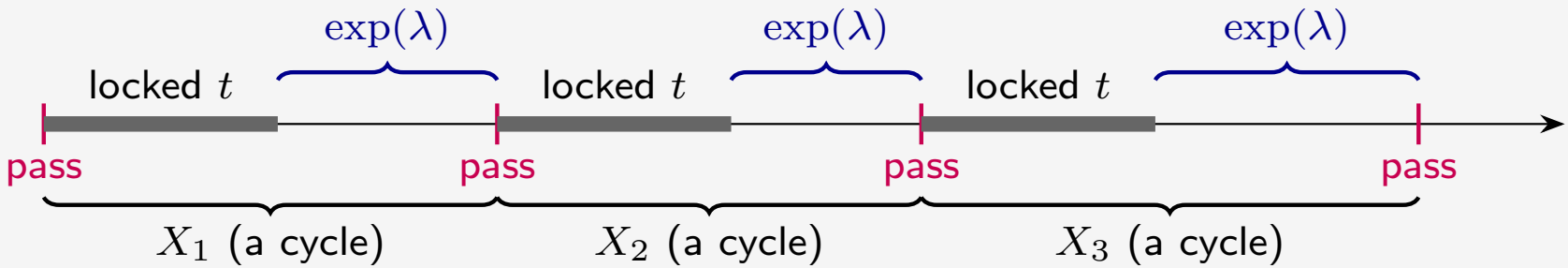
Class problem 9

3. Customers arrive to a single-server queueing system according to $PP(\lambda)$. The arriving customer must pass through a door to reach the server. Each time a customer passes through, the door is locked for the next t units of time. If the door is locked when a customer arrives, the customer is lost incurring a cost c. After passing through the door, if the server is free, the customer gets service. If not, the customer is lost incurring a cost K. If the service times follow $\exp(\mu)$, what is the average cost per unit time?

▶ Colab notebook: CP10 §3 (door with a lockout) `[INCLASS] [full]`

Class problem 9

3. The door: a cycle starts each time a customer passes through



After a customer passes, the door is locked for t; every arrival during the lock is lost (cost c). After the lock opens, the next arrival comes after an $\exp(\lambda)$ time (Poisson arrivals, memoryless) and passes — a new cycle. The cycles are iid, so the passing epochs form a renewal process with

$$X = t + \exp(\lambda), \qquad E[X] = t + \frac{1}{\lambda}.$$

▷ The cost K concerns the server: the passing customer is served if the server is free, lost if it is still busy. Next: the cost of one cycle.

Class problem 9

3. Expected cost per cycle $R = C_1 + C_2$, then divide by $E[X]$

C_1 = cost of the customers lost at the locked door: Poisson(λt) of them, so $E[C_1] = \lambda t\, c$.

C_2 = cost K if the customer passing at the *end* of the cycle finds the server busy. $Y \sim \exp(\mu)$: remaining service time at the cycle start (memoryless — the same whether or not the first customer was served); $Z \sim \exp(\lambda)$: wait after the lock. Lost iff $Y > t + Z$:

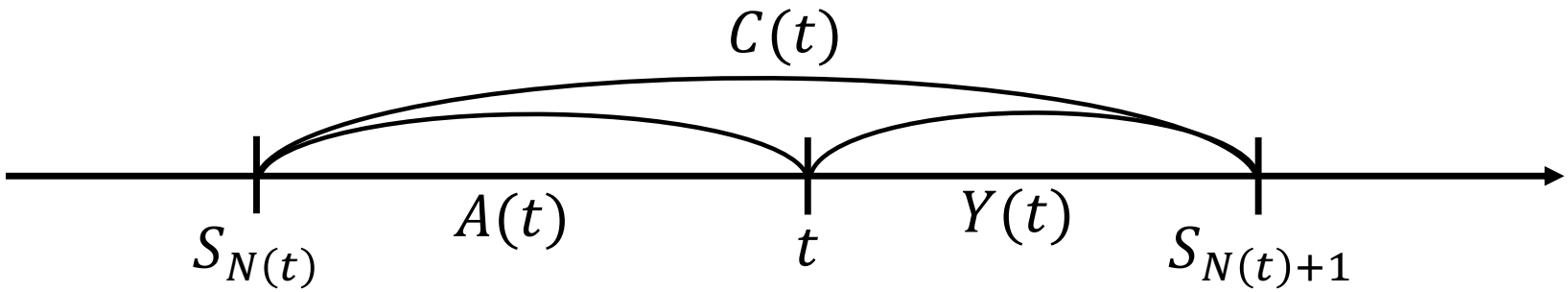
$$E[C_2] = K\, P(Y > t + Z) = K \int_0^\infty \lambda e^{-\lambda z}\, e^{-\mu(t+z)}\, dz$$

$$= K\, e^{-\mu t} \int_0^\infty \lambda e^{-(\lambda+\mu)z}\, dz = \boxed{K\, e^{-\mu t}\, \frac{\lambda}{\lambda+\mu}}\ ,$$

$$\lim_{u\to\infty} \frac{E[R(u)]}{u} = \frac{E[R]}{E[X]} = \boxed{\frac{\lambda t\, c + K\, e^{-\mu t}\lambda/(\lambda+\mu)}{t + 1/\lambda}}\ .$$

$\lambda = 1$/min, $t = 1$ min, $\mu = 0.5$/min, $c = 1$, $K = 5$: $(1 + 2.022)/2 \approx 1.51$ per minute (CP10 §3).

Age and remaining time



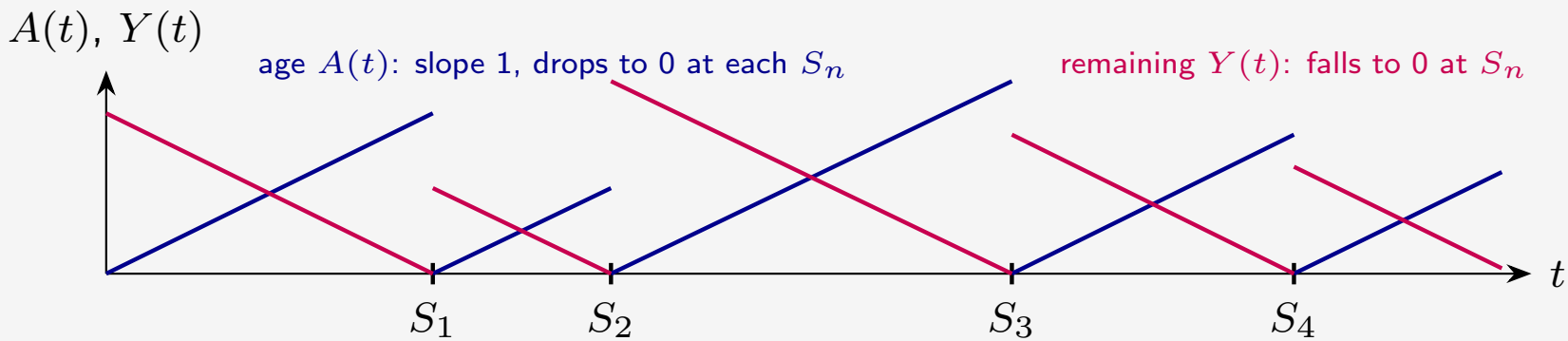
$$C(t) = A(t) + Y(t).$$

▷ Companion §1.6 and Proof 6: the time averages of $A(t)$ and $Y(t)$ are both $E[X^2]/(2E[X])$.

Age and remaining time

Three random variables attached to a time t

$A(t) = t - S_{N(t)}$ (**age**: time since the last renewal), $Y(t) = S_{N(t)+1} - t$ (**remaining** or residual life), $C(t) = X_{N(t)+1} = A(t) + Y(t)$ (the length of the interval that covers t).



Is $X_{N(t)+1} \sim F$? **No**: the interval that covers t is more likely to be a long one — it is sampled by *length* (inspection paradox, next frames).

▷ $A(t)$ itself never converges (a sawtooth); what converges is its *time average* — as the limiting probabilities of Ch. 5 are time fractions.

Age and remaining time

Time average of the age — a renewal reward argument

Reward of cycle n = area under the age curve during that cycle: $R_n = \int_0^{X_n} s\,ds = X_n^2/2$ (iid, each depending on its own X_n). Split the integral at the last renewal before t:

$$\frac{1}{t}\int_0^t A(s)\,ds = \frac{\sum_{n=1}^{N(t)} R_n}{t} + \frac{\int_{S_{N(t)}}^t A(s)\,ds}{t} \xrightarrow[t\to\infty]{} \frac{E[X^2/2]}{E[X]} + 0,$$

the last term being at most $X_{N(t)+1}^2/(2t) \to 0$ (needs $E[X^2] < \infty$). The falling curve has the same area per cycle, and $C = A + Y$ adds up to $E[X^2]/E[X]$:

$$\lim_{t\to\infty}\frac{1}{t}\int_0^t A(s)\,ds = \lim_{t\to\infty}\frac{1}{t}\int_0^t Y(s)\,ds = \frac{E[X^2]}{2E[X]} \quad \text{a.s.}$$

▷ $\exp(\lambda)$: $1/\lambda$; uniform$(0,1)$: $\frac{1}{3}$ (CP10 §B); deterministic μ: $\mu/2$.

Inspection paradox

$$C(t) = A(t) + Y(t).$$

$$\lim_{t\to\infty} \frac{\int_0^t C(s)\,ds}{t} = \lim_{t\to\infty} \frac{\int_0^t A(s)\,ds}{t} + \lim_{t\to\infty} \frac{\int_0^t Y(s)\,ds}{t} = \frac{E[X^2]}{E[X]} \text{ a.s.}$$

$$\frac{E[X^2]}{E[X]} - E[X] = \frac{\mu^2 + \sigma^2}{\mu} - \mu = \frac{\sigma^2}{\mu} > 0.$$

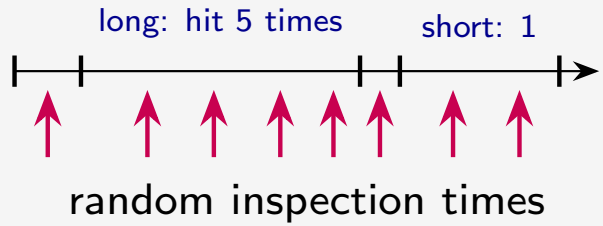
- Intuition: if we choose t equally likely, then it has higher probability to choose t in a longer interval.

▷ Companion Proof 6: the interval covering a random time is length-biased — density $xf(x)/E[X]$.

Inspection paradox

Why $E[\text{length of the interval covering } t] \geq E[X]$: sampling by length

Pick a time t "at random". By renewal reward with $R = X\,\mathbf{1}\{X \in [x, x+dx]\}$, the fraction of the time axis covered by intervals of length in $[x, x+dx]$ is $x\,f(x)\,dx/E[X]$: the interval covering a random time has the length-biased density $xf(x)/E[X]$.



Its mean: $$\int_0^\infty x \cdot \frac{xf(x)}{E[X]}\,dx = \frac{E[X^2]}{E[X]} = E[X] + \frac{\sigma^2}{E[X]} \geq E[X]$$

with equality only for deterministic gaps ($\sigma = 0$).

The same bias, three ways

mean length seen $= E[X^2]/E[X]$ · mean age = mean remaining $= E[X^2]/(2E[X])$ · a Poisson arrival ("random time") waits $E[X^2]/(2E[X])$ for the next renewal, not $E[X]/2$ — the bus stop of §4.

Inspection paradox

In daily life (Scientific American, J. Murtagh, 18 Jan 2024)

Trains. Headways alternate $4, 12, 4, 12, \ldots$ minutes (mean 8). A passenger arriving at a random time lands in a 12-minute gap with probability $12/16 = \frac{3}{4}$: the headway she experiences has mean $\frac{4\cdot 4 + 12\cdot 12}{16} = 10$ min $= E[X^2]/E[X]$, and her mean wait is $E[X^2]/(2E[X]) = 5$ min, not $E[X]/2 = 4$.

Friends. Four people with $3, 2, 1, 2$ friends (mean 2). Averaged over each person's friends, the friends have $1.67,\ 2.5,\ 3,\ 2.5$ friends — mean $2.42 > 2$; counting every friendship from both ends, a friend has on average $\sum_i d_i^2 / \sum_i d_i = 18/8 = 2.25 = E[D^2]/E[D]$ friends: the same length-biased formula with the degree D in place of X. Popular people are counted by more friends, long intervals by more inspection times.

The call centre is always busy when you call — the same reason.

▷ Units selected "by being there" are biased toward large ones.

Regenerative process

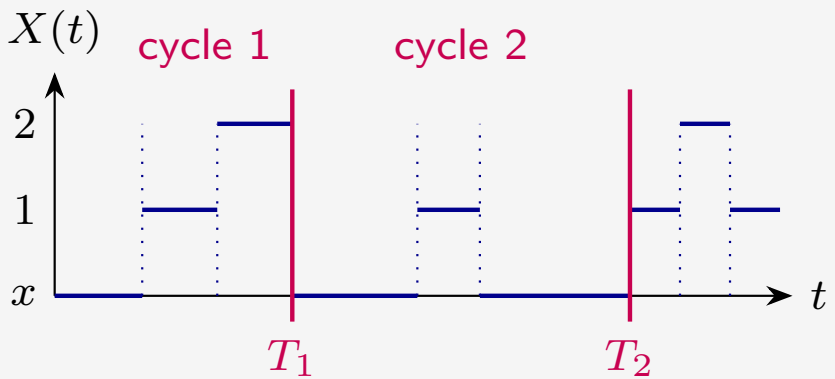
Definition

Let $\{X(t), t \geq 0\}$ be a stochastic process. Whenever $X(t)$ hits some values, let's say x, it is restarted. Then, $\{X(t), t \geq 0\}$ is called a regenerative process.

▷ Companion §1.7 and Proof 7: cycles are iid, so the regeneration epochs form a renewal process.

Regenerative process

Cycles: independent copies of the whole process



Regeneration epochs $T_1 < T_2 < \cdots$: whenever the process hits x it restarts, so the pieces of the path between successive T_n are iid copies and the cycle lengths are iid.

$N(t) = \#\{\text{regenerations in } (0, t]\}$

is a renewal process.

Examples

A renewal process itself (regenerates at each S_n); an ergodic CTMC (at each entry into a fixed state i); a queue (when a customer arrives to an empty system); the on/off system of the alternating renewal process. The Markov property is *not* required — only iid cycles.

Regenerative process

- Suppose $X(\tau_1) = x$, $X(\tau_2) = x$ for $\tau_1 < \tau_2$. Then, $X(\tau_1 + t)$ and $X(\tau_2 + t)$ are stochastically identical.
- If we consider a counting process that counts the number of regenerations until time t, it is a renewal process.
- For a regenerative process, we are interested in the fraction of time that the system stays in state j.
- Using the result of the renewal reward process,

$$\begin{aligned}\text{Fraction of time in state } j &= \frac{E[\text{time in state } j \text{ during a cycle}]}{E[T]} \\ &= \lim_{t\to\infty} P\left(X(t) = j\right).\end{aligned}$$

Regenerative process

Why: renewal reward with "time spent in j" as the reward

Let $R_n =$ time spent in state j during the n-th cycle (iid, each depending on its own cycle) and $T =$ a cycle length. Then

$$\frac{1}{t}\int_0^t \mathbf{1}\{X(s)=j\}\,ds \xrightarrow[t\to\infty]{} \frac{E[\text{time in } j \text{ during a cycle}]}{E[T]} \quad \text{a.s.}$$

is the long-run fraction of time in j — exactly the renewal reward theorem.

Time fraction versus limiting probability

The slide's second equality, $\lim_{t\to\infty} P(X(t)=j) = E[\text{time in } j]/E[T]$, needs a *non-lattice* cycle length (not concentrated on multiples of some $d>0$); for a lattice cycle $P(X(t)=j)$ may oscillate forever while the time fraction still exists. With a cycle density (this course) both agree.

▷ Ch. 5 took p_j from $pQ=0$ and deferred "time fraction" to here.

Example: ergodic CTMC

- We can think that whenever the system visits state i, the system restarts due to Markov property. So,

$$\text{Fraction of time in state } j = \frac{\text{mean time in state } j \text{ during a cycle}}{\text{mean time from } i \text{ to } i}.$$

▷ Companion Proof 7: $p_j = (1/\lambda_j)/m_j$ — the step that Ch. 5 Proof 6(c) deferred to this chapter.

Example: ergodic CTMC

An ergodic CTMC regenerates at every entry into a fixed state i

Markov property + exponential sojourns: at each entry into i the past is forgotten, so the successive $i \to i$ excursions are iid cycles with mean length m_i (the mean return time). For any state j,

$$p_j = \frac{E[\text{time in } j \text{ during an } i\text{-cycle}]}{m_i},$$

$$j = i: \quad p_i = \frac{E[\text{one sojourn in } i]}{m_i} \implies \boxed{p_i = \frac{1/\lambda_i}{m_i}}, \qquad m_i = \frac{1}{\lambda_i p_i}.$$

This is the formula Ch. 5 Proof 6(c) and HW3 A3(v) used ("entries into i occur at rate $\lambda_i p_i$").

Check: machine up/down (CP8 §1)

Cycle = one up time $\exp(\mu)$ + one repair $\exp(\lambda)$, starting at each repair completion: $p_{\text{up}} = \dfrac{1/\mu}{1/\mu + 1/\lambda} = \dfrac{\lambda}{\lambda + \mu}$, as from $pQ = 0$ — but the cycle formula only needs iid cycles, not exponential clocks (HW3 Part C).

Example: alternating renewal process

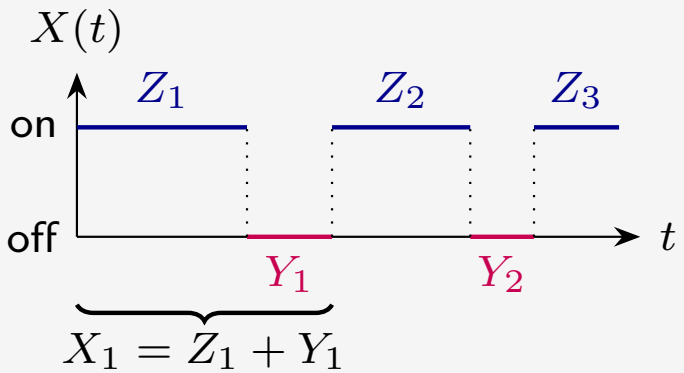
- Consider a system that changes states between on/off
- Z: random variable representing "on times"
- Y: random variable representing "off times"
- Z_n: n^{th} on time
- Y_n: n^{th} off time
- X_n: n^{th} cycle time, $Z_n + Y_n$

$$\text{Fraction of time that the system is on} = \frac{E[Z_n]}{E[X_n]}.$$

▷ Companion Proof 8 (final-style): on-fraction $E[Z]/(E[Z]+E[Y])$; the age process; insensitivity (HW3 Part C explained).

Example: alternating renewal process

On/off cycles



Cycle $X_n = Z_n + Y_n$ (on + off), $E[Z], E[Y] < \infty$; the pairs (Z_n, Y_n) are iid (Z_n, Y_n may depend on each other). Renewal reward with $R_n = Z_n$:

$$\text{fraction of time on} = \frac{E[Z]}{E[Z] + E[Y]}$$

($= \lim_t P(\text{on at } t)$ for a non-lattice cycle).

Examples

Machine with up times of mean m_u and repair times of mean m_r, any shapes: availability $m_u/(m_u + m_r)$. The bank of §2: busy fraction $\mu_G/(\mu_G + 1/\lambda)$. The premium rates of §5: "on" = paying r_1.

Example: alternating renewal process

Final-style: the fraction of time the age is at most c

Call the system *on* at time t if $A(t) \le c$. In a renewal interval of length X the age is $\le c$ for $\min(X, c)$ time units: an alternating renewal process with on-time $Z = \min(X, c)$ and cycle X. The tail formula gives $E[\min(X, c)] = \int_0^\infty P(\min(X, c) > x)\, dx = \int_0^c (1 - F(x))\, dx$, so

$$\text{fraction of time with } A(t) \le c = \frac{\int_0^c (1 - F(x))\, dx}{E[X]} =: F_e(c)\,,$$

the **equilibrium distribution** of F (the same for $Y(t)$, by symmetry of the cycle).

Checks

$F = \exp(\lambda)$: $F_e(c) = \lambda \int_0^c e^{-\lambda x} dx = F(c)$ — the age of a Poisson process is again $\exp(\lambda)$. Uniform$(0, 1)$: $F_e(c) = 2c - c^2$; the age is $\le \frac{1}{4}$ for $43.75\,\%$ of the time although only $25\,\%$ of the intervals are that short (an age is measured *inside* an interval).

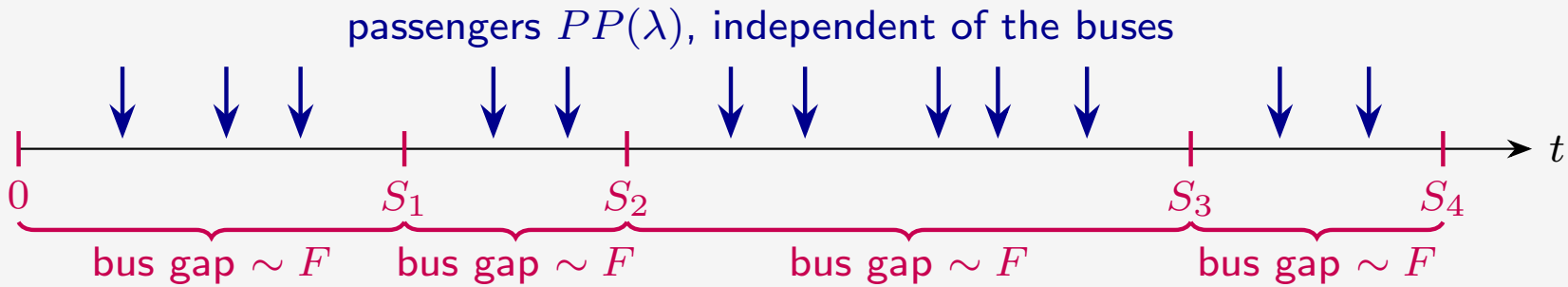
Class problem 9

4. Consider a bus stop. Passengers arrive to the bus stop according to $PP(\lambda)$. Buses arrive according to a renewal process with inter-arrival distribution F. Passengers and buses arrive to the bus stop independently.
   (a) the average number of people at the bus stop in the long run.
   (b) the average amount of time that a passenger waits in the long run.

▶ Colab notebook: CP10 §4 (bus stop) `[INCLASS] [full]`

Class problem 9

4(a). Number waiting: reward = passenger-minutes accumulated in a bus cycle



Buses are the renewal epochs ($X \sim F$); each bus takes everyone waiting. In a cycle the number waiting s time units after the bus is a Poisson count $N_p(s)$ with $E[N_p(s)] = \lambda s$; the reward is $R = \int_0^X N_p(s)\, ds$:

$$E[R] = \int_0^\infty E\Big[\int_0^x N_p(s)\, ds\Big] dF(x) \overset{\text{Fubini}}{=} \int_0^\infty \int_0^x \lambda s\, ds\, dF(x)$$
$$= \int_0^\infty \frac{\lambda x^2}{2}\, dF(x) = \frac{\lambda E[X^2]}{2}$$

$$L = \text{long-run average number waiting} = \frac{E[R]}{E[X]} = \boxed{\frac{\lambda\, E[X^2]}{2E[X]}}$$

Class problem 9

4(b). A passenger's wait: the remaining time seen by a Poisson arrival

Given one passenger arrival in $(0, t]$, its time is uniform on $(0, t)$ (Poisson property), independent of the buses: a passenger arrives at a "random time" of the bus process and waits the *remaining* time until the next bus, so her long-run average wait is the time average of Y:

$$W = \frac{E[X^2]}{2E[X]} \quad \big(\geq E[X]/2\big), \qquad L = \frac{\lambda E[X^2]}{2E[X]} = \lambda W \quad \text{(Little's law, Ch. 8)}.$$

Numbers: $\lambda = 2$ passengers/min, three bus schedules with mean headway 10 min

bus gaps F	$E[X^2]$	$W = E[X^2]/(2E[X])$	$L = \lambda W$
deterministic 10	100	5.0 min	10
uniform$(5, 15)$	108.3	5.42 min	10.8
exponential, mean 10	200	10.0 min	20

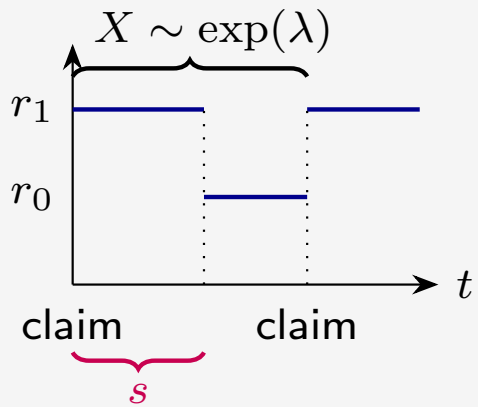
Class problem 9

5. The rate a certain insurance company charges its policyholders (customers) alternates between r_1 and r_0. A new customer is initially charged at a rate r_1 per year. When a customer paying at rate r_1 has made no claims for s consecutive years, the rate becomes r_0 per year. The rate remains at r_0 until a claim is made, at which time it reverts to r_1. If a customer makes claims according to $PP(\lambda)$,
   (a) calculate the proportion of time that the customer pays at rates r_1 and r_0 in the long run.
   (b) obtain the long-run average amount paid per year.

▶ Colab notebook: CP10 §5 (insurance premiums) `[INCLASS] [full]`

Class problem 9

5. Premium rate: cycles from claim to claim



A claim resets the rate to r_1; claims form a $PP(\lambda)$, so the time to the next claim is $\exp(\lambda)$ whatever happened before: **cycle** = claim-to-claim interval $X \sim \exp(\lambda)$, $E[X] = 1/\lambda$. The customer pays r_1 for $\min(s, X)$ and r_0 for $(X - s)^+$:

$$E[\min(s,X)] = \int_0^s x\lambda e^{-\lambda x}dx + se^{-\lambda s} = \tfrac{1}{\lambda}\left(1-e^{-\lambda s}\right).$$

(a) at r_1 : $P_1 = \dfrac{E[\min(s,X)]}{E[X]} = \dfrac{(1-e^{-\lambda s})/\lambda}{1/\lambda} = 1 - e^{-\lambda s}$,

at r_0 : $P_0 = 1 - P_1 = e^{-\lambda s}$;

(b) average per year $= r_0P_0 + r_1P_1 = r_0e^{-\lambda s} + r_1\left(1 - e^{-\lambda s}\right)$.

$\lambda = 0.2$/yr, $s = 3$ yr, $r_1 = 1000$, $r_0 = 600$: $P_0 = e^{-0.6} \approx 0.549$, average ≈ 780.5 per year (CP10 §5).